

Amir Hosein Heydar

Quant Researcher | Systematic Trading Research | Algorithmic Trading Developer
bs.amir.heydar@gmail.com | github.com/bsAmirHeydar | Remote

QUANT RESEARCHER

Junior / Mid-Level Roles

TARGET ROLES

- Quant Researcher
- Quantitative Researcher
- Algorithmic Trading Researcher
- Systematic Trading Researcher
- Quant Developer / Research Engineer
- Trading Systems Developer

CORE STACK

Python
MQL4 / MQL5
MetaTrader 4 / 5
Backtesting
Git / GitHub
Tickstory workflows
TypeScript basics

QUANT FOCUS

Market regime analysis
Trend-following logic
Risk management
Strategy validation
Edge monitoring
Probability & statistics
Decision-making under uncertainty

PUBLIC PORTFOLIO

[GitHub Profile](#)

10 public repositories across Python, MQL5, TypeScript, and software projects.

POSITIONING

Preferred name: Amir.
Independent quant researcher with practical trading experience, systematic strategy development, and real business systems background.

PROFESSIONAL SUMMARY

Independent quant trading researcher and algorithmic trading developer focused on systematic market research, hypothesis-based alpha validation, market regimes, risk management, and explainable trading systems. Started programming early with C++ and Python, later moving into Python-based backtesting, MQL4/MQL5 development, MetaTrader workflows, data-driven strategy testing, and discretionary trading. Current work centers on Decision Alpha Lab: a modular research framework designed to test the assumptions behind trading edges before turning them into strategy logic.

Quant Research

Python/MQL

Backtesting

Regimes

Risk

Systematic Trading

FEATURED RESEARCH PROJECT

Decision Alpha Lab | github.com/bsAmirHeydar/decision-alpha-lab

A Python/MQL quantitative research framework for hypothesis-first systematic trading. The project is built around a simple principle: a trading strategy should not be evaluated only as a black box. Each idea has underlying assumptions about trend persistence, regime behavior, volatility, risk, liquidity, and payoff asymmetry. Decision Alpha Lab separates those assumptions, tests them independently, and then connects validated logic to research, monitoring, visualization, and execution modules.

RESEARCH & TRADING EXPERIENCE

Independent Quant Trading Researcher | Self-employed | Remote

2021 - Present

- Built Python-based backtesting prototypes for indicator, rule-based, and strategy-level research.
- Developed multiple MQL4/MQL5 Expert Advisors, indicators, and MetaTrader trading tools.
- Worked with historical-data workflows, including Tickstory-based data preparation for MT4 research.
- Tested thousands of strategy variations and gained practical experience with debugging, robustness checks, and overfitting limitations.
- Passed multiple prop-firm evaluations and achieved three prop-firm withdrawals while translating discretionary experience into systematic risk logic.
- Built trend-following research logic with positive historical expectancy across different markets, timeframes, and periods, emphasizing robustness and parameter stability.

Business Systems & Growth Operations | Family Business / Self-employed | Iran

- Applied algorithmic and systems-based thinking to ERP-style workflows, marketing operations, and business decision processes.
- Helped build a repeatable growth system that generated 100,000+ followers and thousands of inbound customer leads.

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QUANT RESEARCH VALUE PROPOSITION

My edge as a candidate is the combination of programming, market experience, systematic research, and business-level systems thinking. I am not positioning myself as a signal provider or discretionary trader. I am positioning myself as a research-oriented builder of trading systems, testing workflows, and decision frameworks that make market assumptions measurable and monitorable.

TECHNICAL & RESEARCH CAPABILITIES

Quant Research

- Hypothesis design
- Market regimes
- Trend behavior
- Risk / payoff asymmetry
- Edge monitoring

Engineering

- Python research tools
- MQL4/MQL5 systems
- MetaTrader workflows
- Backtesting
- Git/GitHub portfolio

Execution Mindset

- Robustness over overfit
- Limited downside
- Optionality / convexity
- Drawdown interpretation
- Process monitoring

PUBLIC GITHUB REPOSITORY PORTFOLIO

Portfolio: <https://github.com/bsAmirHeydar>

decision-alpha-lab

Python/MQL research framework for hypothesis-based systematic trading, regimes, risk control, and edge monitoring.

breakout-micro-following

MQL5 breakout-following research system focused on structured entries, execution logic, and market behavior.

pullback-trading

MQL5 pullback trading research system for rule-based entries, order logic, and chart-based validation.

micro-structure

MQL5 market microstructure research tools for imbalance, regime behavior, and execution hypotheses.

volatility-following

MQL5 volatility and fat-tail research tools for regime-aware trading systems.

divergence

MQL5 divergence-based trading research system with session, candle, entry, and order modules.

CLS63

Experimental MQL5 trading system for hook-based entries, risk-controlled execution, and visual review.

Gartal-Inc

Public software / business-system repository demonstrating broader systems and product thinking.

Aura

Public application/software repository demonstrating general development and project-building experience.

next

Public web/software repository demonstrating broader development practice beyond trading systems.